

23 - S2 - Q4

(a) 4 clustering methods + application

(b) HAC : single.

Solution (a)

① (i) Centroids-based clustering

(2) Describe:

Centroids-based clustering is also known as partitioning method. It is a type of clustering that divides the data into non-hierarchical groups. The most common example of partitioning clustering is the k-means clustering algorithm.

(3) Scenario: we roughly know how many clusters (k) we want and the clusters are roughly spherical in feature space.

② (i) Hierarchical clustering

(2) Hierarchical clustering is one of the most commonly used methods for summarizing the property of data structure. A hierarchical tree is a nested set of partition of samples, represented by a tree diagram or dendrogram

(3) Scenario : we don't need to know the number of clusters and want a multi-level view of the clustering structure.

(3.1) Distribution-based clustering

(2) The data is divided based on the probability of how a dataset belongs to a particular distribution.

The grouping is done by assuming some distributions, e.g. Gaussian distribution.

The example of this type is the Gaussian Mixture Models (GMM).

(3) Scenario

When data are well-modeled by parametric distributions, or when we want a probabilistic interpretation

④ Density-based clustering

(2) The Density-based clustering method connects the highly dense areas into clusters, and arbitrary shaped distributions are formed as long as the dense region can be connected.

(3) Scenario: When clusters are of arbitrary shape and you also want to identify outliers/noise. Common in spatial data, image analysis, anomaly detection.

c) ①

$$d_{s_1, s_2} = 0.8137$$

$$d_{s_1, s_3} = 1.7223$$

$$d_{s_1, s_4} = 2.1423$$

$$d_{s_2, s_3} = 2.3113$$

$$d_{s_2, s_4} = 2.4234$$

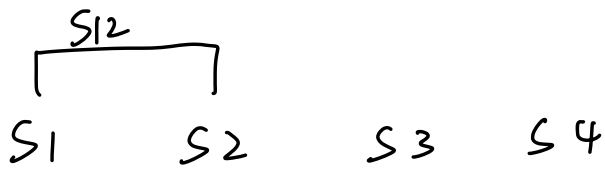
$$d_{s_3, s_4} = 2.4387$$

	s_1	s_2	s_3	s_4
s_1	0			
s_2	0.8137	0		
s_3	1.7223	2.3113	0	
s_4	2.1423	2.4234	2.4387	0

d_{s_1, s_2} is smallest, so we merge them

and denote it as s_{12}

The dendrogram is as follow



$$\textcircled{2} \quad d_{S_{12}, S_3} = \min \{ d_{S_1, S_3}, d_{S_2, S_3} \}$$

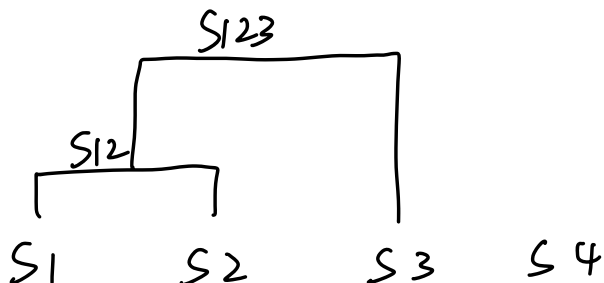
$$= 1.7223$$

$$d_{S_{12}, S_4} = \min \{ d_{S_1, S_4}, d_{S_2, S_4} \}$$

$$= 2.1423$$

	S_{12}	S_3	S_4
S_{12}	0		
S_3	1.7223	0	
S_4	2.1423	2.4387	0

$$\{ S_{12}, S_3 \} \rightarrow S_{123}$$



$$\textcircled{3} \quad d_{S_{123}, S_4} = \min\{d_{S_1, S_4}, d_{S_2, S_4}, d_{S_3, S_4}\} \\ = 1.1423$$

	S_{123}	S_4
S_{123}	0	
S_4	1.1423	0

$$\{S_{123}, S_4\} \rightarrow S_{1234}$$

